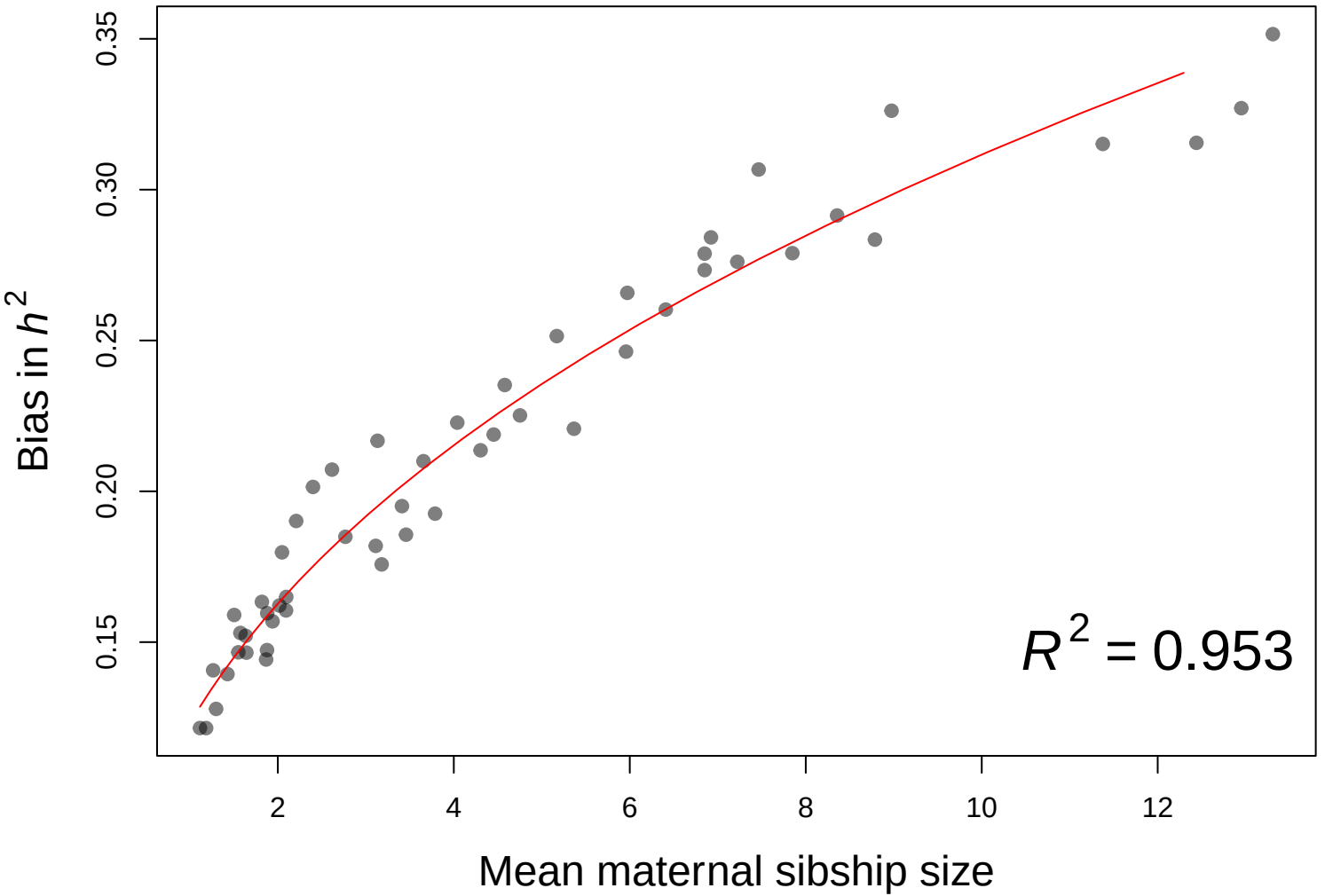
a)

Metric	R^2
records	0.218
maternities	0.391
paternities	0.329
full sibs	0.539
maternal sibs	0.744
maternal half sibs	0.694
paternal sibs	0.681
paternal half sibs	0.644
maternal grandmothers	0.175
maternal grandfathers	0.167
paternal grandmothers	0.138
paternal grandfathers	0.121
maximum pedigree depth	−0.018
founders	−0.006
mean maternal sibship size	0.953
mean paternal sibship size	0.857
non−zero F	0.072
F > 0.125	0.197
mean pairwise relatedness	0.004
pairwise relatedness>=0.125	0.012
pairwise relatedness>=0.25	0.001
pairwise relatedness>=0.5	−0.019
var pairwise relatedness	−0.009

b)